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# Evaluation on the utility efficiency of metro infrastructure projects in China from sustainable development perspective

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**Abstract**

The development of metro infrastructure projects (MIP) has been playing an important role in the economic growth in China through improving urban transportation condition. The economic benefits from MIP in China have been well appreciated. However, concerns on the utility efficiency and the impacts of MIP on social and environmental aspects have been increasingly raised. It is anticipated that a large number of metro infrastructure projects will be built in the near future in China. There is a need for a method to guide the development of MIP towards achieving better utility efficiency and collective benefits between economic, social and environmental dimensions. Previous studies have provided various methods for evaluating economic performance of MIP, but it appears that there is no existing method for studying the utility efficiency of a metro infrastructure project particularly from sustainable development perspective. This paper presents a model for evaluating the utility efficiency of MIP with reference to the Chinese context, which is called utility efficiency evaluation index (UEEI) model. Population of city (POP), length of Metro systems (LEN), annual ridership of Metro systems (RID), ticket price (FAR) and Gross Domestic Product (GDP) are selected as the variables for developing the UEEI model. The utility efficiency of the 17 MIPs is analyzed by using the data collected from 17 cities in China.
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*Keywords:* Metro infrastructure project; Sustainable performance; China; Utility efficiency

## 1. Introduction

Metro infrastructure project (MIP) is a common means in cities for improving transportation condition thus contributing to the urban economic development. Francesca and Enrica (2011) opined that light rail, metro and other urban rail transit systems can play a significant role in improving the attractiveness and quality of urban public transport. There are over 180 cities in the world which have developed the metro system (http://mic-ro.com/metro/). At present, the development of metro system in China is still at the primary stage compared to that in other developed countries. From 1978 to 2010, China's urban population increased from 80 million to about 640 million. By the end of 2010, there are 124 cities in China with a population of more than 1 million (China Statistical Yearbook, 2011). However currently there are only 17 cities which have a metro system in China. Compared with developed countries, for instance, UK, USA, France and others, the development level of metro system in China is still very low in terms of both scale and management performance.

The development of a metro system is a major strategy globally to drive social and economic development. The implementation of metro infrastructure projects has contributed significantly to the development of urbanization in China. The study by Zhou (2004) suggests that the investment of 1 Yuan on metro in Chongqing will lead to the increase of GDP by 2.63 Yuan, and currently the Chongqing Metro has provided 8466 jobs. It is estimated that the number of MIP will continue to grow globally particularly in those developing countries such as China where there are huge plans for urbanization. According

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to "the 12th five year plan transportation system", the total distance of urban rail transit in China will reach 3000 km, and 21 more cities will develop metro systems in the country (National Development and Reform Commission, NDRC, 2012).

The implementation of MIPs involves huge amount of social, financial and environmental resources. Previous studies demonstrate that the installation of metro systems requires large amount of capital investment, generally ranging from US$15 to 300 million per route-km with no guarantee of sound financial returns (Allport, 1990; Halcrow, 2000; Loo and Li, 2006). In the "Shanghai urban rail transit network planning", 210 billion Yuan will be invested to build 880 km of Shanghai urban rail transit, accounting for about 239 million Yuan per route-km (Song and Wang, 2009). Therefore, good performance of an MIP is expected from all stakeholders, including government, public, investor and other professional groups.

However, the performance of MIP varies between different cities and countries due to various reasons such as technical skill, management level and others. For example, New York City Subway, Paris Metro, London Underground, Tokyo Metro, and MTR of Hong Kong are commonly referred to as good examples. Both New York City and Hong Kong have well established rail systems, characterized by an extensive network structure and high transit ridership (Becky et al., 2010). London has the most advanced metro system, which has the first metro line in the history. This metro system is still the biggest one in Europe (Xi, 2008). Similarly, metro in Paris has been used for more than 110 years since its development in early 1900s and this system is well appreciated by the public for its convenience and time-saving (http://www.ccmetro.com/newsite/readnews.aspx?id=267). Tokyo is the first Asian city to use an urban rail transit. According to the study by Shu and Shi (2008) the metro system in Tokyo has made great contribution to the development of the city.

Other studies have examined the performance of metro system from different perspectives. Mohring (1972) pointed out that investments in rail transit infrastructure divert marginal automobile travellers away from their vehicles, resulting in a traffic diversion effect, and thereby reducing air pollution. Huang and Xia (2011) argue that metro system has direct and indirect effect on social and economic development and suggests that urban rail transit has become important for developing urban economy, improving industrial infrastructure and raising people living standard. The research work conducted by Francesca and Enrica (2011) presents the correlation between land-use plan and utility efficiency of urban rail system in a specific city. It emphasizes the proper land plan in order to have better efficiency in operating metro system. In examining the characteristic of city' rail networks, Guo and Zhu (2003) have built a more effective evaluation system by using a quantitative method, and studied the Beijing Metro Line 5 from the perspective of social benefit, the economic benefit and the utility capacity of the metro line. Liu and Hu (2007) investigated the Chongqing metro system, and found that there is a good correlation between real estate price and the location of individual stations of the metro system. The real estate nearby the stations will generally enjoy better price. Wang and Xi (2012) presented an assessment model for examining the benefit of air pollution reduction from the development of metro system. Stokesa and MacDonald, 2008developed a model to assess the potential cost savings in public health from a new light rail transit system in US.

In fact, a metro system is closely associated with people's life and work. The efficiency of using this type of infrastructure plays an important role in contribution to urban sustainable development. Nevertheless, there is little study on the utility efficiency of an urban metro system from a sustainable development perspective. Previous studies focus on studying economic performance of metro infrastructure projects. In fact, the implementation of a metro infrastructure project has significant impacts to social and environmental performance. Litman (2007) pointed out that high quality performance of metro system can reduce urban traffic congestion, and the metro system improvements can be cost effective investments when all economic impacts are considered. Xiao (2013) considers that urban rail transit plays an important role in not only improving the urban transportation contradiction, but also optimizing the use of urban land resources. Jiang and Bai (2010) compared the metro system with other traffic systems, and found obvious advantage from using the metro system in saving energy and reducing the air pollution in cities. Some other studies suggest that metro is "green transport" because of its high speed, punctuality, large carrying-capacity, independence from the ground traffic, use of clean energy, and low energy consumption. Therefore, this kind of traffic system is very important and necessary for implementing the principle of urban sustainable development (Bi, 2008). Although a number of cities in China have introduced a metro system, it appears that their utility efficiency is not good from the perspective of urban sustainable development. Some researchers have pointed out the weak areas in operating metro systems in China compared with other overseas cities. Liu and Chen (2008) found that the proportion of passengers who choose metro as means of transportation in China is less than that in Tokyo, London and Moscow. The urban rail transit in China has experienced more than four decades of development. Nevertheless, many problems are appreciated, for example, the lack of overall consideration on sustainable urban development during design and planning stages, giving the focus only on engineering technologies and network design, the lack of consideration on the future development of rail transit in urban planning, lack of an evaluation system to assess the utility efficiency of metro system and thus not being able to provide information for further improvement of the systems (Li and Xu, 2007). It is therefore considered essential to assess the metro infrastructure project (MIP) utility efficiency that integrates economic, social and environmental horizons from a sustainable perspective. Allport (1990) has pointed early the importance of evaluating the efficiency of metro system from the finding that there is no yardstick to compare the efficiency of metro system at both regional and global level. Metro lines and stations provide passengers more choices in urban areas where there are diversified sectors including business, hospital, education, and others. There is an urgent need for a method that can assist

decision makers or planners to assess the utility performance when deciding to implement an MIP. The aim of this paper therefore is to study a method that enables evaluating the utility efficiency of a metro infrastructure project from a sustainable perspective with particular reference to China.

## 2. Methodology

This research starts with understanding the typical existing methods of evaluating the performance of metro infrastructure projects. These methods are reviewed through examining various literature publications. The principles of these methods are examined and their weaknesses are appreciated from the perspective of utility efficiency, which leads to the proposal of a utility efficiency evaluation index (UEEI) model. The validity of UEEI has also been discussed through interviews. By applying the index model to a sample of 17 cases collected in 17 Chinese cities, the utility efficiency of the MIP in these cities is presented. A scoring model is used to rank the utility performance of metro systems between these 17 cities. In this scoring model, four variables are used, namely, LEN/POP (length of system per million population), RID/POP (annual ridership per million population), RID/LEN (annual ridership per kilometer of system) and FAR/GDP (ratio of fare to per capita GDP). In order to assess the performance score for individual metro systems, benchmarks for these four variables are established. Data for these city cases are collected through visiting relevant public and governmental websites, and these websites are listed in Table 1.

## 3. Previous studies on evaluating the performance of MIP

Efficiency is defined as the quality of doing something well and effectively, without wasting time, money, or energy (The Longman Dictionary). Generally, there are three groups of stakeholders in relation with the efficiency of metro systems, namely, the government, operating companies and passengers. Those three groups have different interests. Government usually prefers to subsidize less to maintain the operation of the system; passengers prefer to pay less to enjoy better service; and the operating company works to maximize the ratio of revenue to cost. But there is little research on the indicators or evaluating model for studying efficiency of metro projects. Many researchers have studies the efficiency of bus transportation system. Bresson and Dargay (2003) used vehicle km/capita, seats km/capita, and frequency on the transportation work density to compare the determinants of the demand for public transportation between England and France. In a study on the impact of management effectiveness on urban rail efficiency, Jain and Cullinane (2008) introduced five technical indicators: rail network length, number of staff in operating company, number of carriages, passenger trips and car-kilometers. Most of the researches on transportation efficiency were conducted from the operating companies' perspective. Sampaio and Neto (2008) selected four indicators to examine the efficiency of public transportation systems including operational cost, number of equivalent vehicles, number of employees in operating company, and number of passengers transported as indicators. Yu (2008) presented the benchmarks for evaluating technical efficiency of bus system by using five indicators, including number of bus, fuel consumption, network length, service area population and passenger kilometers. In recent years, the concerns on passengers have attracted more attention. A study by Guo and Nigel (2011) evaluated the performance of metro systems from the passenger perspective through investigating the extra costs due to transfer inconvenience.

However, little research has been conducted on the transportation efficiency from the sustainable development perspective. The work by Niu (2007) presents a quantitative model for evaluating sustainable performance of metro systems. This model is developed by using fuzzy principle to calculate the sustainability of urban rail transit projects. In this fuzzy principle based model, there are four sub-models: social influence, economic benefit, environmental impact and management performance. Three attributes are used to measure the performance of the four sub-models, namely, development status, the level of harmony, and level of sustainability. However, there is no method in Niu's model for calculating the values of these three attributes, which presents the limitation for the application of the model. Furthermore, Lei (2006) presents a model for estimating the sustainability performance of Chinese metro. The model incorporates four variables, namely, transportation capacity, saving ability for transportation resources, environmental and ecological protection, and profitability. However, the use of this model requests for defining weighting values between the variables, but the method for defining the weights is not discussed.

Some other researchers introduced effective indicators to evaluate the performance of metro system, but there is little discussion on how to measure these indicators quantitatively. Xiao (2013) evaluated the performance of metro system by referring to eight aspects, including social, economy, ecology, technology, policy, management, city and the traffic. Similarly, Liu and Chen (2008) proposed the following eight aspects for evaluating the sustainable performance of metro system, including consultation with planning, system designing, operation management, consumer satisfaction, post-evaluation on

Table 1
The websites used for data collection.

| Website | Description |
|---|---|
| http://wenku.baidu.com/view/4e01ad619b6648d7c1c746d2.htmlhttp://tieba.baidu.com/p/2084905940 | Website used for the collection of the population in 17 Chinese cities |
| http://wenku.baidu.com/view/8acd6916ff00bed5b9f31d27.html | Website used for the collection of the GDP in the 17 cities |
| http://blog.hangzhou.com.cn/archiver/?tid-9231426.html | Website used for the collection of the ticket price of 17 metro systems |
| http://wenku.baidu.com/view/310718462b160b4e767fcf61.html | Website used for the collection of the ridership of 17 metro systems |

environmental effects, economic analysis, and degree of public participation.

There are still other scholars who have conducted research in the discipline of sustainable transportation. Dai and Chen (2004) introduced a model for assessing the performance of sustainable transportation in Shanghai. The AHP method is applied in selecting indicators establishment of the assessment model. Chen and Liao (2009) appreciated the characteristics of highway transportation, identified the factors, offering the sustainability performance of projects, and then established the sustainability evaluation index system by composing the attributes of economic effect, social effect, resource degradation, environmental influence and development ability.

From the above discussion, the typical indicators applied in previous studies for evaluating the performance of metro infrastructure projects can be summarized in Table 2. It can also be noted that the previous studies do not provide a model for conducting the evaluation of utility efficiency of metro system.

## 4. Utility efficiency evaluation index model for metro infrastructure project (MIP)

The indicators or factors affecting the utility efficiency of metro projects include those factors addressed in the previous section, which are associated with metro projects. It is appreciated that other external factors such as parking lots in the downtown area may affect the metro project efficiency. However, it is considered proper to focus on project related parameters in examining their utility efficiency.

In Table 2, some indicators are measureable quantitatively, for example, GDP and number of passenger. And some others are not measureable, for example, passenger comfort and passenger satisfaction. On the other hand, the data collection for some indicators is very difficult if not possible, for example, rate of empty loading (non-passenger) and operation income particularly in the Chinese context. Accordingly, five indicators are selected for building up the utility efficiency evaluation index model, including population of city (POP), length of Metro systems (LEN), annual ridership of Metro systems (RID), ticket price (FAR) and Gross Domestic Product (GDP). The data of these five indicators can be collected from various information sources, such as the website of http://www.ditiezu.com/. However, the values of these indicators vary significantly between different cities because of different characteristics embodied in individual cities, such as population scale, economic development level, and the history of metro system development. Therefore, it is considered not proper to compare the values of these indicators between different cities. In order to eliminate the effects of individual city characteristics on the comparison study, relative measures are considered more proper. By examining the meaning and relationship of the five indicators (LEN, POP, RID, FAR, GDP), the following four relative index measures are established:

$$\alpha_1 = \frac{\mathbf{LEN}}{\mathbf{POP}} \quad \text{(length of system per million population)}$$

$$\alpha_2 = \frac{\mathbf{RID}}{\mathbf{POP}} \quad \text{(annual ridership per million population)}$$

$$\alpha_3 = \frac{\mathbf{RID}}{\mathbf{LEN}} \quad \text{(annual ridership per kilometer of system)}$$

$$\alpha_4 = \frac{\mathbf{FAR}}{\mathbf{GDP}} \quad \text{(ratio of fare to per capita GDP)}$$

The parameters $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$ collectively form a model, called utility efficiency evaluation index model (UEEI). This model can also be presented in a structured model, as shown in Fig. 1. These four indexes reflect the utility efficiency of metro projects from four dimensions.

It is considered that the four indexes can effectively reflect the utility efficiency of metro system. The parameter $\alpha_1$ denotes the length of system per million population. Traditionally, the total metro system length is used to tell the level of metro development in a city (Yu, 2008). However, it is considered that the total length cannot truly reflect the level of metro development to a given city as different cities have different characteristics such as population size. Therefore, it is more effective to indicate the development level by per capital length. A larger $\alpha_1$ indicates both larger metro scale and better service for the public by providing the access for passengers to get their destination. In other words, a larger $\alpha_1$ will give more opportunity to passengers who will use metro system, which indicates a better utility efficiency.

The parameter $\alpha_2$ means the annual ridership per million population. This index can reflect the ratio of how many times

Table 2
Typical indicators for performance evaluation on MIPs.

| Indicator | Author |
|---|---|
| Roadnetwork density, Number of passenger | Chen and Liao (2009) |
| Network length, Service area population, Employees | Yu (2008) |
| City population, GDP, Per capita GDP, Traveltime<br>Rate of empty loading, Rate of empty loading<br>Per capita rail transit length | Dai and Chen (2004) |
| Operating costs, Operation income, Impact on social development, Impact on the vicinity life,<br>Impact on regional economy | Xiao (2013) |
| Passenger comfort, Passenger satisfaction,<br>Passenger flow estimation, Appropriateness of metro planning, Appropriateness of metro design | Liu and Chen (2008) |
| Number of train cars, Trips and car-kilometers, Number of staff | Jain and Cullinane, 2008 |
| Vehicle km/capita, Seats km/capita, Frequency on the network | Bresson and Dargay, 2003 |

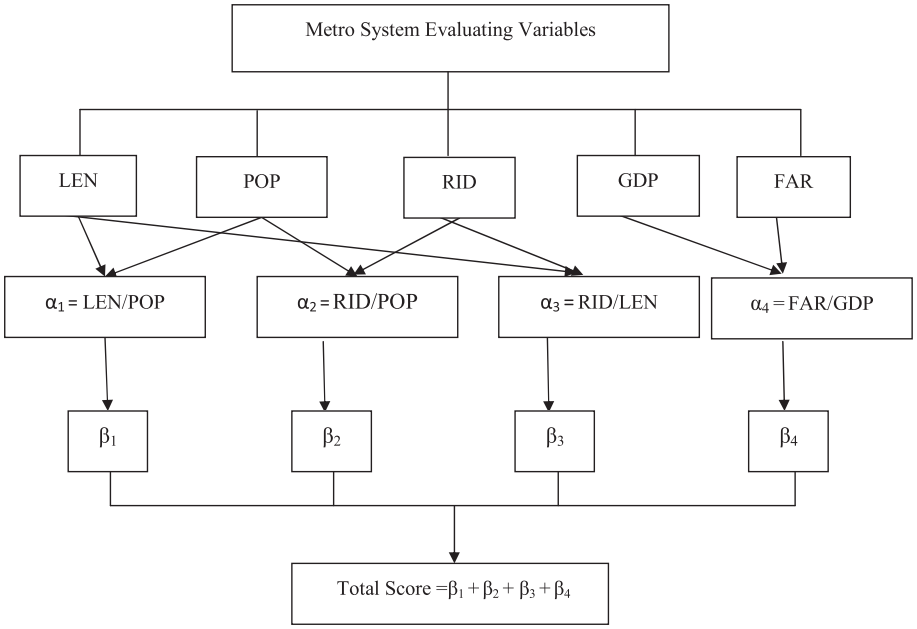



Fig. 1. Utility efficiency evaluation index model.

in average that people in a city use metro system per year. As the majority of metro systems users are commuters, the ridership in weekdays is much larger than that in weekend. Ridership on different lines or stations distributes unevenly because different lines have different surrounding destinations and the number of passengers varies in different time periods. Thus the annual ridership is considered more effective to reflect the number of how many people use the metro system in the whole year. A lager $\alpha_2$ reveals to a certain extent the dependencies of citizens on Metro system, which represents the attraction of the metro network to citizens. The higher $\alpha_2$ may also explain that the management level of metro system is better. From the perspective of sustainable development, a higher $\alpha_2$ is desirable because more people use metro and thus less use cars and other vehicles; consequently, the energy for cars and other vehicles will be reduced.

The parameter $\alpha_3$ means the annual ridership per kilometer of metro. This index tells the intensity of transit which reflects directly the utility efficiency of metro system. A proper level of $\alpha_3$ is important. If this value is too big, it means that passengers suffer from crowding, although this may bring better revenues. In this situation, the operation company should improve their service, for example, increasing the transit frequency, and this way will still allow for raising the return of their investment. On the other hand, a lower $\alpha_3$ will reduce the revenue whilst the operation cost will not change much. Therefore, it is important to maintain a proper level of $\alpha_3$.

The parameter $\alpha_4$ denotes the ratio of fare to per capita GDP. This index reflects passenger's affordability. This is the key factor influencing the use of metro system by the public. Cheaper fare encourages more people to choose metro, especially the low income groups. For example, in the Beijing metro system, fare reduction adjustment in 2006 led to the increase of the rate of citizens using metro by 15% in the following year (http://www.bjpc.gov.cn/). As living standards vary a lot in different cities in China, it is not reasonable to apply same fare scale as the affordability is different in different cities. To mitigate this weakness, the ratio $\alpha_4$ is considered effective in presenting comparable affordability. On the other hand, the size of GDP in a city cannot represent people's affordability because different cities have different population scales. Therefore, per capita GDP adjusted by purchasing power parity can represent people's affordability.

Whilst the evaluation index model UEEI is primarily developed from literature review and analysis of utility efficiency of metro projects, the validity and efficiency of the model (UEEI) have been further discussed with five experts with interview. These five interviewees include three professors and two managers from Chongqing Metro Operation Firms. All interviewees have knowledge of metro operation. It is generally agreed between the interviewees that the index model UEEI is effective and applicable.

## 5. Data collection and analysis

This section is to present the values of the four indexes by using the data collected from 17 Chinese cities. By examining relevant websites, the research data have been collected as shown in Table 3.

By applying the data in Table 3 to the four indexes $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$, the values of these indexes are calculated, as shown in Table 4. The original value of FAR is too small compared to the size of per capita GDP. For convenience in calculating, $\alpha_4$ values in Table 4 are the results of amplifying the original value of FAR by 10,000 times.

Table 3
Research data collected from 17 cities.

| City | LEN (km) | POP (million) | RID (million) | Per Capital GDP (Yuan) | FAR (Yuan per 10 km) |
|---|---|---|---|---|---|
| Beijing | 442 | 7.30 | 2781.34 | 104,675.6 | 1.25 |
| Shanghai | 439.1 | 9.50 | 2445.5 | 79,878.44 | 1.77 |
| Guangzhou | 235.9 | 4.15 | 2117 | 82,654.82 | 2.37 |
| Shenzhen | 178.9 | 1.06 | 839.5 | 58,702.62 | 2.24 |
| Tianjin | 139.4 | 5.07 | 156.95 | 88,661.44 | 1.65 |
| Chongqing | 130.4 | 3.93 | 401.5 | 57,841.94 | 1.99 |
| Nanjing | 84.6 | 2.82 | 401.5 | 114,193.3 | 0.95 |
| Dalian | 64.45 | 2.12 | 43.80 | 51,323.44 | 1.26 |
| Wuhan | 56.9 | 4.49 | 219.00 | 46,814.99 | 1.41 |
| Shenyang | 41 | 3.98 | 62.05 | 92,781.36 | 1.61 |
| Changchun | 49 | 2.22 | N/A | 104,675.6 | 1.66 |
| Hangzhou | 48 | 1.93 | 58.40 | 79,878.44 | 1.88 |
| Chengdu | 41 | 2.34 | 200.75 | 82,654.82 | 2.44 |
| Suzhou | 25 | 1.17 | 40.15 | 58,702.62 | 2.40 |
| Xi'an | 20.5 | 2.59 | 73.00 | 88,661.44 | 2.44 |
| Kunming | 18 | 1.55 | 4.00 | 57,841.94 | 2.78 |
| Foshan | 14.8 | 0.43 | 40.15 | 114,193.3 | 3.38 |

Further analysis is given to the correlation between the indexes, $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$.

As there are no data available about ridership of metro in Changchun city, this paper used the other 16 cities' data to investigate the correlation between the four indexes. Based on the data in Table 4, Excel 2007 of Microsoft Office and IMB SPSS 19.0 are used to calculate the correlation values. And the results are shown in Table 5.

From Table 5, it can be noted that there is a strong positive correlation between $\alpha_1$ and $\alpha_2$.This indicates that a larger metro length per million population will promote more people to use the metro system. This further demonstrates that the parameters $\alpha_1$ and $\alpha_2$ are effective to reflect the utility efficiency of metro system. Furthermore, there is also a significant positive correlation between $\alpha_2$ and $\alpha_3$. It indicates that when people use more metro (a higher $\alpha_2$), the ridership per kilometer ($\alpha_3$) will be more highly improved. In other words, the function of providing public transportation by metro system is better realized, thus the utility efficiency is improved.

Table 4
The calculation results of $\alpha_1$, $\alpha_2$, $\alpha_3$, and $\alpha_4$.

| City | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ |
|---|---|---|---|---|
| Beijing | 60.57 | 104.43 | 6.29 | 0.14 |
| Shanghai | 46.24 | 70.56 | 5.57 | 0.21 |
| Guangzhou | 56.78 | 139.60 | 8.97 | 0.22 |
| Shenzhen | 169.01 | 217.28 | 4.69 | 0.22 |
| Tianjin | 27.52 | 8.49 | 1.13 | 0.17 |
| Chongqing | 33.14 | 27.96 | 3.08 | 0.51 |
| Nanjing | 29.98 | 38.98 | 4.75 | 0.11 |
| Dalian | 30.43 | 5.67 | 0.68 | 0.12 |
| Wuhan | 12.68 | 13.37 | 3.85 | 0.18 |
| Shenyang | 10.30 | 4.27 | 1.51 | 0.19 |
| Changchun | 22.04 | N/A | N/A | 0.28 |
| Hangzhou | 24.84 | 8.28 | 1.22 | 0.21 |
| Chengdu | 17.51 | 23.49 | 4.90 | 0.42 |
| Suzhou | 21.36 | 9.40 | 1.61 | 0.21 |
| Xi'an | 7.92 | 7.73 | 3.56 | 0.48 |
| Kunming | 11.62 | 0.71 | 0.22 | 0.59 |
| Foshan | 34.46 | 25.62 | 2.71 | 0.36 |

Table 5
Correlation analysis of $\alpha_1$, $\alpha_2$, $\alpha_3$, and $\alpha_4$.

| | | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ |
|---|---|---|---|---|---|
| **$\alpha_1$** | Correlation | 1 | .926** | 0.395 | −0.228 |
| | Sig. (2-tailed) | | 0 | 0.13 | 0.395 |
| | N | 16 | 16 | 16 | 16 |
| **$\alpha_2$** | Correlation | .926** | 1 | .686** | −0.232 |
| | Sig. (2-tailed) | 0 | | 0.003 | 0.388 |
| | N | 16 | 16 | 16 | 16 |
| **$\alpha_3$** | Correlation | 0.395 | .686** | 1 | −0.185 |
| | Sig. (2-tailed) | 0.13 | 0.003 | | 0.493 |
| | N | 16 | 16 | 16 | 16 |
| **$\alpha_4$** | Correlation | −0.228 | 16 | −0.185 | 1 |
| | Sig. (2-tailed) | 0.395 | −0.232 | 0.493 | |
| | N | 16 | 0.388 | 16 | 16 |

** . Correlation is significant at the 0.01 level (2-tailed).

## 6. Discussion

The relative performance of $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$ between these 17 cities can be further discussed in this section. In order to judge the relative performance, it is necessary to establish a benchmark first for these four parameters $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$. The study by Shen and Jiao (2013) collected relevant data on 180 metro systems globally and found that the average values of $\alpha_1$, $\alpha_2$ and $\alpha_3$ between these 180 metro systems are 145.47, 313.44 and 8.88 respectively. By referring to this finding in examining the Chinese cities, it appears that the values of $\alpha_1$, $\alpha_2$, and $\alpha_3$ in the 17 Chinese cities are very low. It shows that metro systems in all Chinese cities are still in the early development stage. Therefore, it is suggested to adopt the maximum value for $\alpha_1$, $\alpha_2$, and $\alpha_3$ among these 17 cities as the benchmarks to $\alpha_1$, $\alpha_2$, and $\alpha_3$ respectively. Accordingly, by referring to Table 4, the benchmarks of these three parameters can be found as follows:

$$\alpha_1^0 = \max_{i=1}^{17}\alpha_{1i} = 169.01 \quad \text{Shenzhen}$$

$$\alpha_2^0 = \max_{i=1}^{17}\alpha_{2i} = 217.28 \quad (\text{Shenzhen})$$

$$\alpha_3^0 = \max_{i=1}^{17}\alpha_{3i} = 8.97 \quad (\text{Guangzhou})$$

As $\alpha_4$ indicates the people's affordability of using metro system, it is considered that the benchmark value of this variable is at a reasonable level, namely, neither too high nor too low. However, there is no reference for establishing the benchmark for the parameter $\alpha_4$. Through reference about the fare data from a report of UBS, 2009, the average value of $\alpha_4$ between the 17 cities is considered effective for its benchmark, which can be written as follows:

$$\alpha_4^0 = \frac{\sum_{i=1}^{17}\alpha_{4i}}{17} = 0.27$$

Therefore the benchmarks for $\alpha_1$, $\alpha_2$, $\alpha_3$ and $\alpha_4$ are as follows:

$$\alpha_1^0 = 169.01;\ \alpha_2^0 = 217.28\ ;\ \alpha_3^0 = 8.97;\ \alpha_4^0 = 0.27$$

By using the above benchmarks, the relative performance between these 17 cities can be calculated by using the following relative performance measures:

$$\beta_1 = 100 - \frac{\left|\alpha_1^0 - \alpha_1\right|}{\left|\alpha_1^0 - \alpha_{1(min)}\right|} \times 100\ (0 \le \beta_1 \le 100)$$

$$\beta_2 = 100 - \frac{\left|\alpha_2^0 - \alpha_2\right|}{\left|\alpha_2^0 - \alpha_{2(min)}\right|} \times 100\ (0 \le \beta_2 \le 100)$$

$$\beta_3 = 100 - \frac{\left|\alpha_3^0 - \alpha_3\right|}{\left|\alpha_3^0 - \alpha_{3(min)}\right|} \times 100\ (0 \le \beta_3 \le 100)$$

$$\beta_4 = 100 - \frac{\left|\alpha_4^0 - \alpha_4\right|}{max\left\{\left|\alpha_4^0 - \alpha_{4(min)}\right|, \left|\alpha_4^0 - \alpha_{4(max)}\right|\right\}} \times 100\ (0 \le \beta_4 \le 100)$$

Where $\alpha_{(min)}$ denotes the minimum value of the parameter $\alpha$ between the 17 cities and $\alpha_{(max)}$ denotes the maximum value of the parameter $\alpha$ between the 17 cities.

As the parameters $\beta_1$, $\beta_2$, $\beta_3$ and $\beta_4$ assume for a minimum value of 0 and a maximum of 100. The higher values of the parameters indicate better utility performance in the corresponding areas. The parameter $\beta_1$ denotes the relative level of the metro length per million population. The parameter $\beta_2$ denotes the relative level of the annual ridership per million population. The parameter $\beta_3$ denotes the relative level of the annual ridership per kilometer of metro. And the parameter $\beta_4$ denotes the relative level of ratio of fare to per capita GDP.

The calculation results of $\beta_1$, $\beta_2$, $\beta_3$, and $\beta_4$ are shown in Table 6.

By using the data in Table 6, the total score for each city can be calculated, and the relative ranks of utility efficiency of metro system between cities can be established according to the total score. Because as noted in the previous section there are no data available about the ridership in the city Changchun, there is no value for $\alpha_2$, $\alpha_3$, $\beta_2$ and $\beta_3$ for Changchun city. Therefore, the calculation for the total score for individual cities will exclude the city Changchun. Accordingly, ranks of relative utility performance are established between 16 cities, as shown in Table 7.

If individual parameters are examined, the utility efficiency performance between cities will be different. For example, if $\beta_1$ is referred, the top five cities will be Shenzhen, Beijing, Guangzhou, Shanghai and Foshan. When $\beta_2$ is referred, the top five cities will be Shenzhen, Beijing, Guangzhou, Shanghai and Nanjing. When $\beta_3$ is referred, the top five cities will be Beijing, Guangzhou, Shanghai, Chengdu and Nanjing. And when $\beta_4$ is referred, the top five cities will be Shenzhen, Suzhou, Guangzhou, Hangzhou and Shanghai. Furthermore, some other individual cities are also good in individual attributes. For example, the metro in Wuhan gives a score of 41.49 for $\alpha_3$, whilst it is not among the top five metro systems. Meanwhile, the metro in Shenyang gives a score of 75 for $\alpha_4$, whilst the total score of Shenyang metro system just ranks 12th in 16 cities.

So, it is considered more proper to adopt an integrated model to evaluate the utility efficiency of metro. The information in Table 7 tells that the top five are metro systems in Shenzhen, Beijing, Guangzhou, Shanghai and Nanjing. The metro performance of total utility efficiency in these cities is obviously better than that in other cities. This is echoed by another report (http://forum.home.news.cn/thread/78303994/1.html, which indicates that the utility performance of Shenzhen, Beijing and Shanghai metro systems has better performance. In fact, the five metro systems in Shenzhen, Beijing, Guangzhou, Shanghai and Nanjing are the top metros in scale in China. According to the statistics in 2013 (http://www.ditiezu.com/), the total length of metro systems in Beijing, Shanghai, Guangzhou and Shenzhen is 442 km, 439.1 km, 235.9 km and 178.9 km respectively. Especially, the total length of metros in Beijing and Shanghai ranks 1st and 2nd globally. These cities have a much better

Table 6
Relative performance.

| No | City | $\beta_1$ | $\beta_2$ | $\beta_3$ | $\beta_4$ |
|---|---|---|---|---|---|
| 1 | Beijing | 32.68 | 47.89 | 69.37 | 59.38 |
| 2 | Shanghai | 23.79 | 32.25 | 61.14 | 81.25 |
| 3 | Guangzhou | 30.33 | 64.13 | 100.00 | 84.38 |
| 4 | Shenzhen | 100.00 | 100.00 | 51.09 | 84.38 |
| 5 | Tianjin | 12.17 | 3.59 | 10.40 | 68.75 |
| 6 | Chongqing | 15.66 | 12.58 | 32.69 | 25.00 |
| 7 | Nanjing | 13.69 | 17.67 | 51.77 | 50.00 |
| 8 | Dalian | 13.97 | 2.29 | 5.26 | 53.13 |
| 9 | Wuhan | 2.95 | 5.85 | 41.49 | 71.88 |
| 10 | Shenyang | 1.48 | 1.64 | 14.74 | 75.00 |
| 11 | Changchun | 8.77 | N/A | N/A | 96.88 |
| 12 | Hangzhou | 10.50 | 3.50 | 11.43 | 81.25 |
| 13 | Chengdu | 5.95 | 10.52 | 53.49 | 53.13 |
| 14 | Suzhou | 8.34 | 4.01 | 15.89 | 81.25 |
| 15 | Xi'an | 0.00 | 3.24 | 38.17 | 34.38 |
| 16 | Kunming | 2.30 | 0.00 | 0.00 | 0.00 |
| 17 | Foshan | 16.48 | 11.50 | 28.46 | 71.88 |

Table 7
The total score of 16 cities.

| Ranks | City | Total Score ($\beta_1 + \beta_2 + \beta_3 + \beta_4$) |
|---|---|---|
| 1 | Shenzhen | 335.46 |
| 2 | Guangzhou | 278.84 |
| 3 | Beijing | 209.32 |
| 4 | Shanghai | 198.43 |
| 5 | Nanjing | 133.14 |
| 6 | Foshan | 128.31 |
| 7 | Wuhan | 122.16 |
| 8 | Chengdu | 123.08 |
| 9 | Suzhou | 109.49 |
| 10 | Hangzhou | 106.68 |
| 11 | Tianjin | 94.91 |
| 12 | Shenyang | 92.86 |
| 13 | Chongqing | 85.92 |
| 14 | Dalian | 74.65 |
| 15 | Xi'an | 75.79 |
| 16 | Kunming | 2.30 |

economic development level compared with other Chinese cities and they have a longer history in developing and operating metro systems. The first line of metro system in Beijing, Shanghai, Guangzhou, Shenzhen and Nanjing was developed in 1969, 1995, 1997, 2004 and 2005 respectively (http://mic-ro.com/metro/). The management mechanism of these systems is more mature and consequently has better utility efficiency.

Furthermore, it is appreciated that the application index factors of UEEI model may change because of the development of cities. For example, considering the tradeoff between utility efficiency of metro ridership and crowdedness, the factors of "crowdedness" should be given more attention when a city is in a more developed stage. In other words, the application of UEEI model should incorporate the dynamic nature of a city development.

The model UEEI provides a theoretical tool for metro project stakeholders, such as planners, investors and project managers, to foresee the expected utility efficiency for those newly planned metro projects. The adjustments on design parameters can be made in order to improve the efficiency value if it is not at expected level. At practical level, UEEI model helps the project stakeholders particularly the project manager to understand the level of utility efficiency of a particular metro system. And solution or decision can be formulated to improve the efficiency through various methods such as renovation, proper maintenance, the application of advanced technology and management systems.

## 7. Conclusion

This study suggests that China will experience a continuous and rapid development in metro system in the coming years. This indicates that a huge amount of resources will be invested on the development of the metro system in future China. Accordingly, it is imperative to ensure that better efficiency can be achieved from the investment. This paper examines the major factors influencing the utility efficiency of metro system, and presents an evaluation mechanism for assessing the utility efficiency. These main factors include the length of metro (LEN), the city population (POP), the annual ridership (RID), per capita GDP of the city (GDP) and the ticket price (FAR). There is a strong correlation between urban economic developments and metro systems. Urban metro system plays an important role in activating and leading regional economic development. The utility efficiency performance between the surveyed 17 cities in China varies significantly. According to the overall performance evaluation, Beijing, Shanghai, Shenzhen, Guangzhou and Nanjing are the top five cities in metro utility performance. However, if individual attributes are applied, the utility efficiency performance of metro system in individual cities will be different. An individual city may be good in one attribute but poor in other attributes. For example, Suzhou is good in FAR/GDP but poor in RID/POP. It demonstrates that the metro system in Suzhou has a proper level of ticket price, but may not be properly designed for considering the population distribution along the metro lines. It signals that the development of further metro systems should consider overall performance particularly at the project planning stage.

Whilst the utility efficiency evaluation index (UEEI) model introduced in this paper is from examining the utility performance of metro systems in the Chinese context, the principle of the model can be applied in examining the metro utility efficiency in other countries. In other words, this study provides a reference methodology for studying the utility efficiency of metro systems in general.

The limitation of this paper is appreciated that the UEEI model does not fully incorporate external factors such as environmental dimension. The lack of this consideration may affect the adequacy of evaluation results on the utility efficiency of metro systems if the evaluation is requested from a holistic view. It is also appreciated that the application of the model has not been tested by real cases, and this issue is currently under study by the research team.

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